

EE6320 RF Integrated Circuits

Project: LNA Design

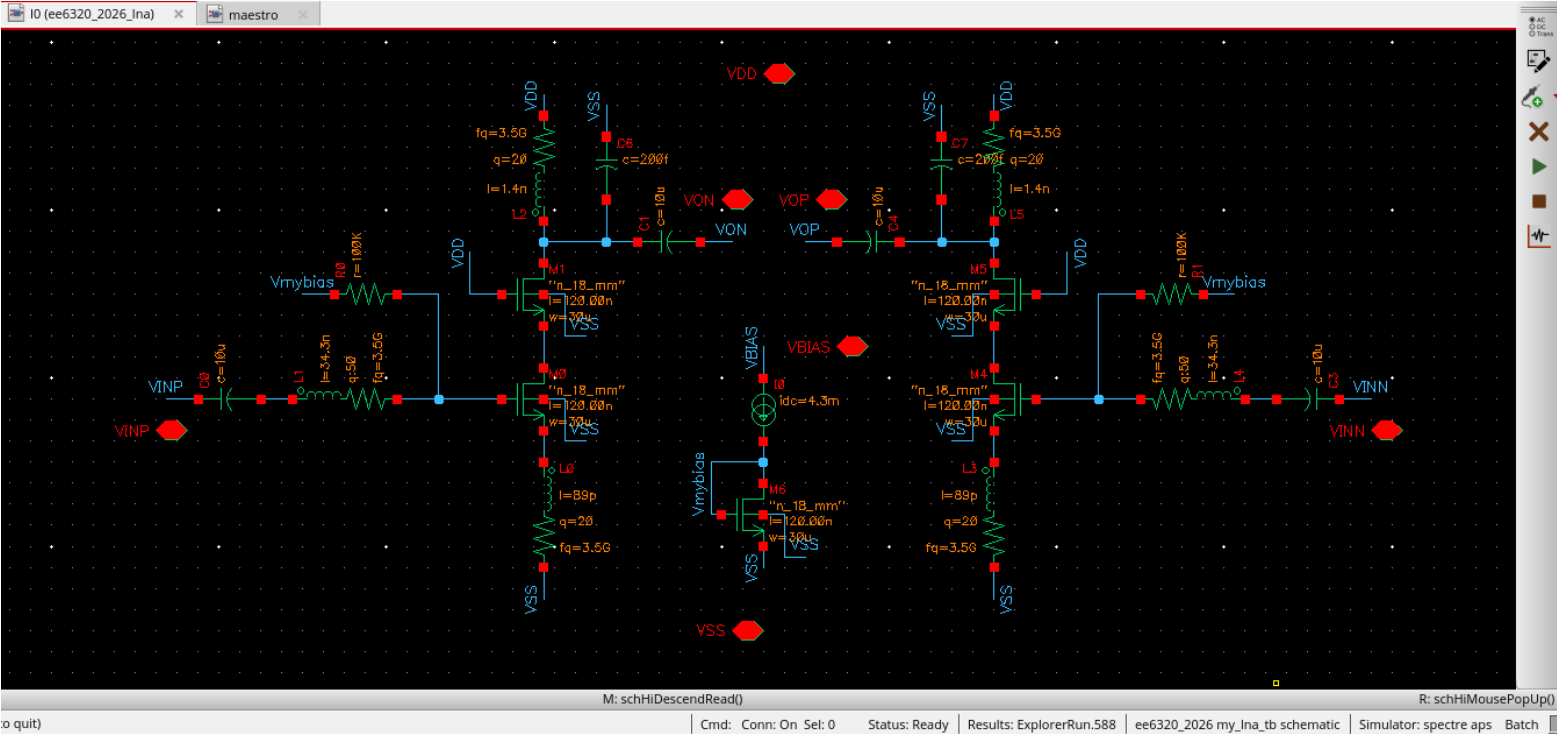
Performance Summary Table

Design metric	Measurement	Simulation Result	Requirement
Input matching	Worst case S11 in the specified band	-12.99dB	< -10dB
	Band over which $S_{11} \leq -10\text{dB}$	3425 to 3580 MHz	3450 to 3550 MHz
Voltage Gain	Minimum Gain in the specified band	21.54dB	$\geq 20\text{dB}$
	Maximum Gain in the specified band	21.83dB	$\geq 20\text{dB}$
	Gain flatness in specified band [Max-Min Gain]	0.29dB	$\leq 1\text{dB}$
	3dB Bandwidth	468 MHz	-
	Load Capacitance [Single Ended]	200f F	200f F
Noise Figure	Maximum Noise Figure in the specified band	1.826dB	$\leq 2\text{dB}$
	Minimum Noise Figure in the specified band	1.793dB	-
	Band over which $NF \leq 3\text{dB}$	2533 to 4486 MHz	-
Linearity	IIP3 Tones used	3.5G Hz, 3.501G Hz	-
Power	Input power used for extrapolation	-50dBm	-
	Power of Fundamental Tone at output (at chosen input power)	-28.185dBm	-
	Power of IM3 Tone at output (at chosen input power)	-108.647dBm	-
	Extrapolated IIP3 LNA DC power consumption [Excluding Bias] Bias circuit power consumption (power supplied by i_bias)	-9.77dBm 8.7996mW 5.16mW	$\geq -10\text{ dBm}$ Minimize Minimize
Other	Sum of all on-chip inductances	2.978nH	-
	Sum of all off-chip inductances (2 Lg's)	68.6nH	-
	Sum of all resistances [Including bias]	200kOhm	-

	Sum of all capacitances [Including AC coupling, excluding load]	40uF	-
	Simulator Used	Spectre	-

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LNA Schematic



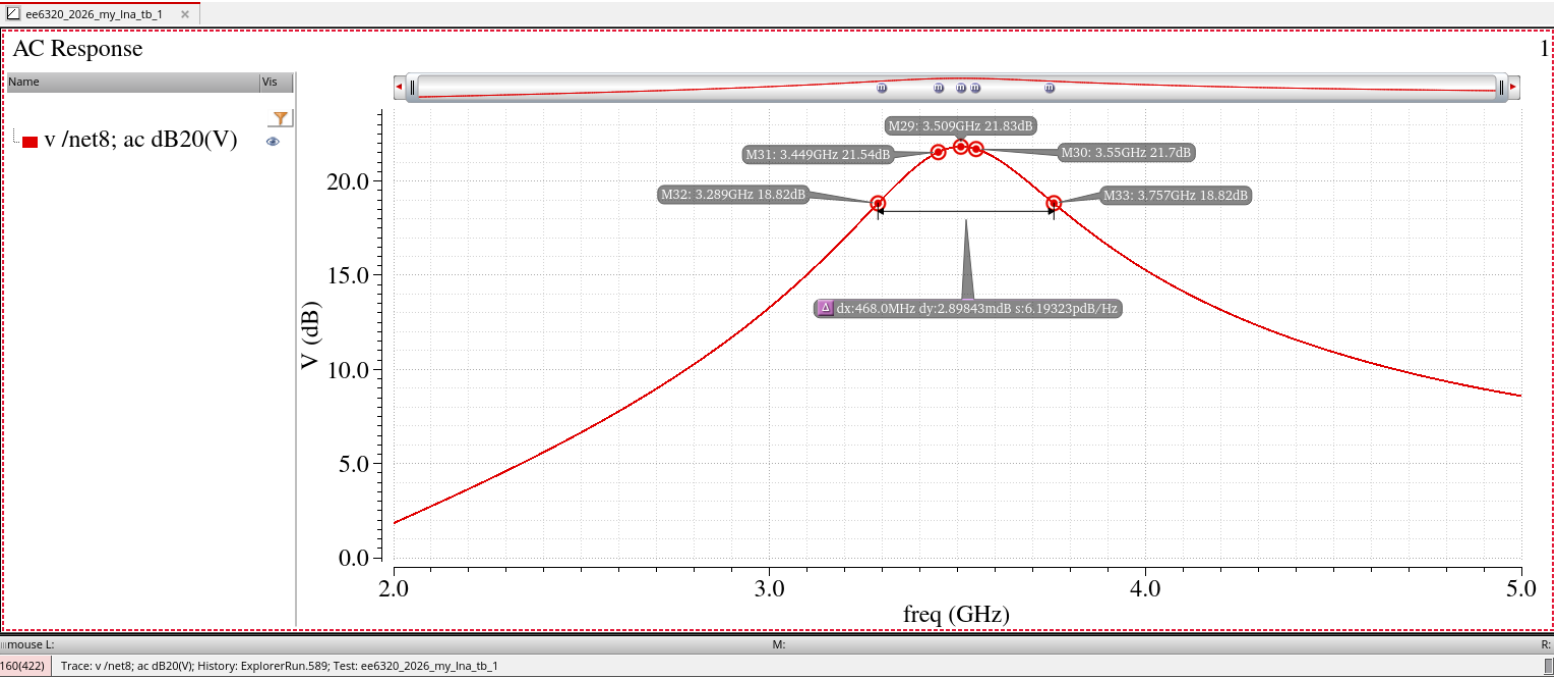
Component Values (one side values)

Design Component	Hand Calculated Value	Simulated Value
Lg (gate inductance)	79.73 nH	34.3 nH
Ls (source inductance)	46.8 pH	89 pH
Ld (drain inductance)	10.2 nH	1.4 nH
Rd* (drain resistance)	4.49 kΩ	0.62 kΩ
R_Is* (res. parallel with Ls)	20.58 Ω	39.1 Ω

Rg* (res. parallel with Lg)	87.67 kΩ	37.71 kΩ
<u>Fixed Constant Parameters</u>		
• LNA MOS parameters: W = 30um, L = 0.12um		
• Current Mirror MOS parameters: W = 30um, L = 0.12um		
• I_bias (current): 4.3mA		
• C_coupling: 10uF		

* - intrinsic resistance to respective inductors (doesn't come in net resistance on/off the chip)

Gain Plot (from AC)



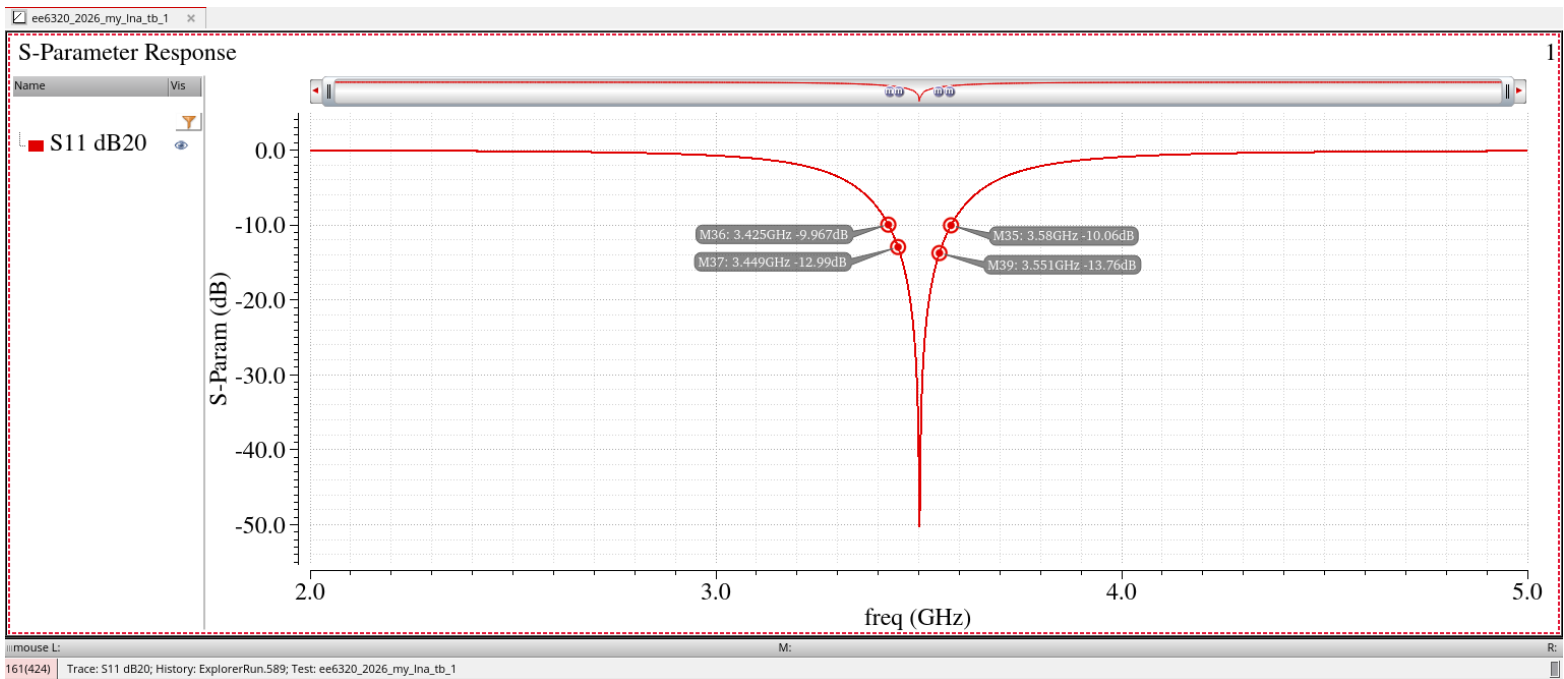
Max Gain through hand calculation: 49.52dB

Max Gain from the simulation: 21.83dB

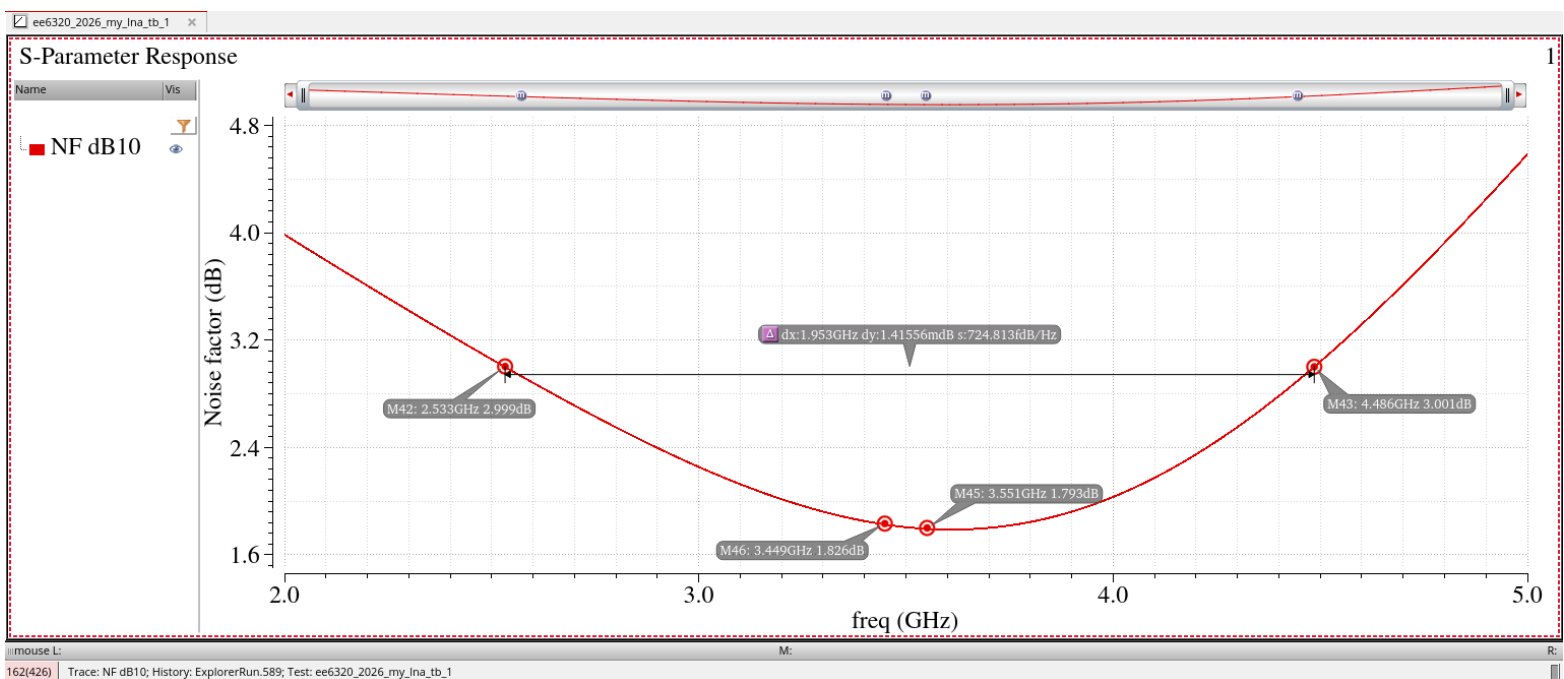
-3dB Gain comes out to be: 18.83dB

From the figure, 3dB bandwidth comes out to be: 468 MHz

S11 Plot



NF Plot



From hand calculations, NF comes out to be: 0.002dB

From the plot, NF comes out to be: 1.826dB

Comments

- The current drawn (both as I_{bias}) and from VDD seem quite high compared to what is expected.
- However, no specifications (S11, NF, Gain and IIP3) seem to be overdesigned for, that is, they just meet the expected specification.
- This suggests that no significant further improvement can be achieved with respect to the power consumption, because there is nothing to offer in the tradeoff.
- However, the factor that limited the minimum current the most was the IIP3 and not NF or gain. This suggests that in case there exists a point for a lower bias current where the non-linearity of g_m (and thus the IIP3) is lesser than the point that has been suggested, there could be a significant improvement in the bias current, without affecting NF or gain much.
- We also observe that the hand-calculated values do not match those of the simulations. The reasons could be –
 - $L_D - C_D$ was assumed to be equal to C_{load} , but that might not be the case if the order of the parasitics is similar to that of C_{load} , due to which the actual L_D needed would be lower.
 - C_{gs} turns out to be about double of the hand calculated value.
 - g_m turns out to be lesser than the hand calculated value. This could be a result of the current in the MOSFETs not being equal to I_{bias} (due to channel length modulation), or due to velocity saturation, or due to the effect of the non-linearity of g_m .
 - L_G – The value of C_{gs} comes out to be about double of what was expected from the hand calculations, so the L_g needed would be around half of what was calculated.
 - L_S – The ω_T from the simulation is smaller than that of the calculations, so the actual L_S needed would be bigger (ignoring the effect of the resistors parallel to L_G and L_S).
- Both the hand calculations of the NF and gain are off by a huge margin from the actual values of the simulation, even when the values for g_m and C_{gs} from the simulations (DC operating point) are plugged in.

This may be due to the quality factors of the inductances L_g and L_s , which end up changing the overall quality factor that causes the upfront gain in the LNA.

It may also be due to the inaccuracy of the noise model of the MOSFETs or the deviation of the MOSFETs in the simulator from the square law.

Hand Calculations done for the Project

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5.3.25 LNA Hand Calculations
($f_0 = 3.5 \text{ GHz}$, $R_s = 50 \Omega$)

For IGM $0.13 \mu\text{m}$, $x_{ox} = 32 \text{ \AA}$
 $\therefore C_{ox} = \frac{\epsilon_{ox}}{x_{ox}} = \frac{3.45 \times 10^{-11}}{32 \times 10^{-10}} = 0.0108 \text{ F/m}^2$

From $\frac{W}{L} = \frac{0.42}{0.15}$, $\beta = 10^{-3}$ from simulation
 $\therefore \mu_n = \frac{\beta \times \frac{W}{L}}{C_{ox}} = 0.033 \text{ m}^2/\text{Vs}$

For $W = 30 \mu\text{m}$, $L = 0.12 \mu\text{m}$, $I_D = 4.3 \text{ mA}$
 $\hookrightarrow$ Assuming $C_0 \approx C_{load} = 200 \text{ fF}$,
 $L_0 = \frac{1}{(2\pi f_0)^2 C_0} \approx 10.2 \text{ nH}$

$\hookrightarrow g_m = \sqrt{2 \mu_n \cdot \text{len} \frac{W}{L} I_D} = 27.68 \text{ mS}$
 $C_{gs} = \frac{2}{3} W L C_{ox} = 0.02592 \times 10^{-12} = 25.92 \text{ fF}$
 $\omega_T = g_m / C_{gs} = 1.068 \times 10^{12} \text{ rad/s}$

$\hookrightarrow L_s = 50 / \omega_T = 46.8 \text{ fH}$

$\hookrightarrow L_s + L_0 = \frac{1}{4\pi^2 f_0^2 C_{gs}} = 79.77 \text{ nH} \Rightarrow L_s = 79.73 \text{ nH}$

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$\alpha_{para} = Q \times (2\pi f_0) \times L$
 $\hookrightarrow \alpha_D = 20 \times (2\pi f_0) \times 10.2 \times 10^{-9} = 4.49 \text{ K}\Omega$
 $\hookrightarrow \alpha_{L_s} = 20 \times (2\pi f_0) \times 46.8 \times 10^{-12} = 20.58 \text{ }\Omega$
 $\hookrightarrow \alpha_g = 50 \times (2\pi f_0) \times 79.73 \times 10^{-9} = 87.67 \text{ K}\Omega$

$V_o = A = -g_m \alpha_D = -1 \times \omega_T \times \alpha_D$
 $V_s = \frac{2}{2\pi f_0 \alpha_g}$
 $|A| = 299.16$ (for $\alpha_D = 515 \text{ }\Omega$)
 $20 \log_{10} A = 49.52 \text{ dB}$

$F = 1 + \gamma g_m \alpha_g (\omega_0 / \omega_T)^2 + 4 \alpha_s / \alpha_D (\omega_0 / \omega_T)^2$
 For $\gamma = \frac{2}{3}$, $\alpha_D = 515 \text{ }\Omega$ $\therefore F = 1.00053$
 $\therefore NF = 10 \log_{10} F = 0.0023 \text{ dB}$

From simulation: $g_m = 17.55 \text{ mS}$, $C_{gs} = 56.7 \text{ fF}$
 $\omega_T = g_m / C_{gs} = 0.31 \times 10^{12} \text{ rad/s}$
 $\therefore |A| = 86.83 \Rightarrow 20 \log_{10} |A| = 38.77 \text{ dB}$
 $\therefore F = 1.0046 \Rightarrow 10 \log_{10} F = NF = 0.02 \text{ dB}$

Design Procedure

- After choosing a suitable starting W_n and I_D (based on estimations for NF and gain using the formulae), we see whether the minimum source resistance (plot of $\text{Real}\{Z_m\}$ in SP) is lesser than 50Ω when L_s is set to 0. If it is not, then we will not be able to achieve a single ended resistance of 50Ω with any appropriate L_s , so we need to choose a more suitable (higher) I_D .
- Then, for that W_n and I_D , we calculate the expected values of L_d , L_s and L_g and plug them in the simulation.
- Then, we tune L_d such that S_{11} has a single negative peak (minima).
- Then, we tune L_g such that the minimum S_{11} occurs at the centre frequency, 3.5 GHz .
- Next, we find the L_s for which $\text{Re}(Z_m)$ is around 50Ω for the best possible matching.
- Then, we obtain the plots for the NF and gain. If either doesn't meet the specification, we can increase our bias current till they start saturating (or decreasing) for any further increase in current. This presumably happens due to the velocity saturation of gm_1 , after which the change in the overall gm (and thus the ω_T) is decided based on the curve of gm_3 vs I_{bias} .
- In the case that this happens, we move to either a higher or a lower W_n , depending on which one helps our specifications and again vary the current to meet our NF and gain specifications; while constantly moving to a different W_n in case the aforementioned saturation of NF and gain with the bias current occurs.
- After a suitable NF and gain have been obtained, we simulate the IIP3. In case it does

not meet the specification, we have to increase the bias current such that we reach a point on the curve of gm_3 vs I_{bias} , where the IIP3 specification is met.

- If the minimum IIP3 still does not meet the specifications, then we move to another appropriate W_n and increase the bias current finely until the IIP3 specification is met.
- During all of this, care should be taken to keep the MOSFETs in region 2 (saturation) by constantly looking at the DC operating point tab for the MOSFETS.
- After all specifications are met, optimization may be performed for the bias current. Moving to a lower W_n causes the bias current for which the IIP3 is minimum to decrease, without noticeably affecting the NF and gain.

Path to the library

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